

ORF 309 — Slides Relevant to Homework 3, Question 4

Compiled from Slides 05, 06, 07

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1 Homework 3, Question 4 — The Problem

Q.4 (Love triangles). A dating website works as follows. There are n people signed up. For every pair of people, the website independently flips a coin that comes up heads with probability p . If the coin comes up heads, the website creates a match between this pair of people. (Being a scam, the site does not actually use any specific information about the people in deciding who to hook up.)

In this setting, it is perfectly possible that the website randomly creates a *love triangle*: three people who are all mutually hooked up.

- a) What is the expected number of hookups a given individual has at once?
- b) What is the probability that a given individual has more than one hookup?
- c) What is the expected number of love triangles on the entire site?

Key ideas needed to solve each part:

- **Part (a):** Indicator random variables + linearity of expectation. Each potential partner gives an independent Bernoulli trial; sum the indicators.
- **Part (b):** The number of hookups for a given person follows $\text{Binom}(n-1, p)$; use the Binomial PMF to compute $P\{\text{more than 1 hookup}\} = 1 - P\{0\} - P\{1\}$.
- **Part (c):** Use indicator random variables over all $\binom{n}{3}$ triples; apply linearity of expectation and independence of edges.

2 Discrete Random Variables and Expectations (Slides 05)

2.1 Types of Random Variables

Definition 2.1 (Discrete Random Variable). A random variable $X : \Omega \rightarrow D$ is called **discrete** if the value set D is a finite or countable set.

Definition 2.2 (Continuous Random Variable). A random variable $X : \Omega \rightarrow D$ is called **continuous** if the value set D is NOT countable.

2.2 Expectation

Definition 2.3 (Expectation of a Discrete Random Variable). If $X : \Omega \rightarrow D$ is a discrete r.v. (DRV) and $f : D \rightarrow \mathbb{R}$ is a value function, then the **expectation** of $f(X)$ is defined as

$$E[f(X)] = \sum_{i \in D} f(i) P\{X = i\}.$$

If $D \subset \mathbb{R}$, we can choose $f(i) = i$ and the expectation of X becomes:

$$E[X] = \sum_{i \in D} i P\{X = i\}.$$

Remark 2.4. This definition makes sense only for DRVs, because for a continuous r.v. (CRV) we have $P\{X = x\} = 0$ for all $x \in D$.

2.3 The Indicator Function (Very Important!)

Definition 2.5 (Indicator Random Variable). Assume A is an event. Then we can define the DRV $\mathbf{1}_A$ with values:

$$\mathbf{1}_A(\omega) = \begin{cases} 1, & \text{if } \omega \in A \\ 0, & \text{if } \omega \notin A \end{cases}$$

The DRV $\mathbf{1}_A$ is called the **indicator random variable** for event A .

Theorem 2.6 (Expectation of the Indicator).

$$E[\mathbf{1}_A] = 1 \cdot P\{\mathbf{1}_A = 1\} + 0 \cdot P\{\mathbf{1}_A = 0\} = P\{A\}.$$

That is, the **expectation of $\mathbf{1}_A$ equals the probability of event A** .

2.4 Usefulness of the Indicator Function

The indicator function allows us to represent the number of successes in a series of trials as a simple sum.

Example 2.7 (Coin Flips). Consider the DRV X representing the number of heads in n coin flips.

- Define the events: $A_i = \{\text{the } i\text{-th flip is heads}\}$ for $i = 1, 2, \dots, n$.
- For each event A_i , define the indicator function $\mathbf{1}_{A_i}$.
- Then we can express X as the sum of indicators:

$$X = \mathbf{1}_{A_1} + \mathbf{1}_{A_2} + \dots + \mathbf{1}_{A_n}.$$

This representation is the key trick for computing expectations of counts.

2.5 Distributions of DRVs

Definition 2.8 (Distribution). If $X : \Omega \rightarrow D$ is a DRV, then its **distribution** is given by specifying the probabilities of all its outcomes $\{P\{X = i\}\}_{i \in D}$.

Definition 2.9 (Uniform Distribution). Let D be a finite set. The DRV $X : \Omega \rightarrow D$ for which

$$P\{X = i\} = \frac{1}{|D|}, \quad \forall i \in D$$

is said to have the **Uniform Distribution** on D . Notation: $X \sim \text{Unif}(D)$.

2.6 Joint and Marginal Distributions

Definition 2.10 (Joint Distribution). The **joint distribution** of two DRVs $X : \Omega \rightarrow D_X$ and $Y : \Omega \rightarrow D_Y$ is specified by the probabilities

$$\{P\{X = i, Y = j\}\}_{i \in D_X, j \in D_Y}.$$

Definition 2.11 (Marginal Distribution). The distribution of DRV X only is called the **marginal distribution** of X .

Claim 2.12 (Recovering Marginals from Joint).

$$P\{X = i\} = \sum_{j \in D_Y} P\{X = i, Y = j\}.$$

Proof sketch: Write $\Omega = \bigcup_{j \in D_Y} \{Y = j\}$ (mutually exclusive partition), so $\{X = i\} = \bigcup_{j \in D_Y} \{X = i, Y = j\}$, and take probabilities.

2.7 Linearity of Expectation

Theorem 2.13 (Linearity of Expectation). For any two DRVs X and Y (no independence required):

$$E[X + Y] = E[X] + E[Y].$$

Proof.

$$\begin{aligned}
E[X + Y] &= \sum_{x \in D_X} \sum_{y \in D_Y} (x + y) P\{X = x, Y = y\} \\
&= \sum_{x \in D_X} \sum_{y \in D_Y} x P\{X = x, Y = y\} + \sum_{x \in D_X} \sum_{y \in D_Y} y P\{X = x, Y = y\} \\
&= \sum_{x \in D_X} x \underbrace{\sum_{y \in D_Y} P\{X = x, Y = y\}}_{=P\{X=x\}} + \sum_{y \in D_Y} y \underbrace{\sum_{x \in D_X} P\{X = x, Y = y\}}_{=P\{Y=y\}} \\
&= E[X] + E[Y].
\end{aligned}$$

□

Remark 2.14. By induction, linearity extends to any finite number of random variables:

$$E[X_1 + X_2 + \cdots + X_n] = E[X_1] + E[X_2] + \cdots + E[X_n].$$

This holds regardless of whether the X_i are independent.

2.8 Indicator Function Properties (Useful Identities)

Exercise 2.15. Prove the following properties of indicator functions:

$$\mathbf{1}_{A \cap B} = \mathbf{1}_A \cdot \mathbf{1}_B \tag{1}$$

$$\mathbf{1}_{A \cup B} = \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_{A \cap B} \tag{2}$$

$$\mathbf{1}_{A^c} = 1 - \mathbf{1}_A \tag{3}$$

3 Independence, Indicator Functions, and Conditional Expectation (Slides 06)

3.1 Indicator Function and Expectation: The Key Example

Example 3.1 (n Coin Flips — Proof that $E[X] = np$) **Setup:** *Flip n identical coins. Let X = total number of heads.*

Define events $A_i = \{\text{the } i\text{-th coin shows H}\}$, with $P\{A_i\} = p$ and $P\{A_i^c\} = 1 - p$.

Define indicator $\mathbf{1}_{A_i}$ for each A_i .

Then $X = \mathbf{1}_{A_1} + \mathbf{1}_{A_2} + \cdots + \mathbf{1}_{A_n}$.

By linearity of expectation and $E[\mathbf{1}_{A_i}] = P\{A_i\} = p$:

$$E[X] = E[\mathbf{1}_{A_1}] + E[\mathbf{1}_{A_2}] + \cdots + E[\mathbf{1}_{A_n}] = p + p + \cdots + p = np.$$

Key remark: We did *not* need to assume A_i and A_j are independent for any $i \neq j$. Even if the coins were correlated (e.g., through collusion), the expected total heads is still np . Independence is not required for linearity of expectation.

3.2 Independence of DRVs

Definition 3.2 (Independence of Two DRVs). Two DRVs $X : \Omega \rightarrow D_X$ and $Y : \Omega \rightarrow D_Y$ are called **independent** ($X \perp Y$) if for every $x \in D_X$ and $y \in D_Y$:

$$P\{X = x \mid Y = y\} = P\{X = x\} \iff P\{X = x, Y = y\} = P\{X = x\} \cdot P\{Y = y\}.$$

Remark 3.3. If $X \perp Y$, then their marginal distributions *determine* their joint distribution.

Theorem 3.4 (Key Rules). For any DRVs X and Y :

- $E[X + Y] = E[X] + E[Y]$ (**always**, no independence required).
- $E[\varphi_1(X) \cdot \varphi_2(Y)] = E[\varphi_1(X)] \cdot E[\varphi_2(Y)]$ (**only if** $X \perp Y$).

Definition 3.5 (Independence of n DRVs). The n DRVs $X_1, X_2, \dots, X_n$ are called **independent** iff for all subsets $\{j_1, j_2, \dots, j_k\} \subseteq \{1, \dots, n\}$ and all values:

$$P\{X_{j_1} = x_{j_1}, X_{j_2} = x_{j_2}, \dots, X_{j_k} = x_{j_k}\} = P\{X_{j_1} = x_{j_1}\} \cdot P\{X_{j_2} = x_{j_2}\} \cdots P\{X_{j_k} = x_{j_k}\}.$$

3.3 Conditional Expectation

Definition 3.6 (Conditional Expectation). If $X : \Omega \rightarrow D_X$ and $Y : \Omega \rightarrow D_Y$ are DRVs and $\varphi : D_X \rightarrow \mathbb{R}$, then the **conditional expectation** of $\varphi(X)$ given $Y = y$ is

$$E[\varphi(X) \mid Y = y] = \sum_{x \in D_X} \varphi(x) P\{X = x \mid Y = y\}.$$

Example 3.7 (Die Rolling Until a 6). **Setup:**

- Roll a fair die repeatedly; stop when we see a 6.
- T = the trial number when the first 6 appears.
- X = the number of 1s seen before the game ends.
- Find: $E[X \mid T = k]$.

Define Z_i = outcome of the i -th trial, with indicator $\mathbf{1}_{\{Z_i=1\}}$.

If the game ends at trial k , then $X = \sum_{i=1}^{k-1} \mathbf{1}_{\{Z_i=1\}}$, so:

$$E[X \mid T = k] = \sum_{i=1}^{k-1} P\{Z_i = 1 \mid T = k\}.$$

Using independence of trials and $\{T = k\} = \{Z_1 \neq 6, \dots, Z_{k-1} \neq 6, Z_k = 6\}$:

$$P\{Z_i = 1 \mid T = k\} = \frac{P\{Z_i = 1, Z_i \neq 6\}}{P\{Z_i \neq 6\}} = \frac{1/6}{5/6} = \frac{1}{5}, \quad \text{for all } i = 1, \dots, k-1.$$

Therefore:

$$E[X \mid T = k] = \sum_{i=1}^{k-1} \frac{1}{5} = \frac{k-1}{5}.$$

4 Bernoulli Processes and the Binomial Distribution (Slides 07)

4.1 Bernoulli Random Variables and Processes

Definition 4.1 (Bernoulli Random Variable). Any random variable $X : \Omega \rightarrow \{0, 1\}$ is called a **Bernoulli RV**.

- $X = 1$ is a “success”; $X = 0$ is a “failure”.
- The indicator function $\mathbf{1}_A$ is a Bernoulli RV for every event A .

Definition 4.2 (Bernoulli Process). A **Bernoulli Process** is an infinite sequence $X_1, X_2, X_3, \dots$ of i.i.d. Bernoulli RVs with

$$P\{X_i = 1\} = p \quad \text{and} \quad P\{X_i = 0\} = 1 - p =: q.$$

4.2 Counting Successes

Definition 4.3 (Counting RV S_n). Define

$$S_n = X_1 + X_2 + \dots + X_n,$$

the **number of successes** in n trials. Each $X_i \in \{0, 1\}$ with $P\{X_i = 1\} = p$, $P\{X_i = 0\} = q$, and $p + q = 1$.

4.3 Deriving the Binomial Distribution

Example 4.4 (Exactly 2 Successes in 8 Trials). We want $P\{S_8 = 2\}$.

- Each specific sequence of 2 successes and 6 failures (e.g., $ssffff$) has probability p^2q^6 by independence.
- The number of such sequences is $\binom{8}{2} = 28$ (choose which 2 of the 8 slots are successes).
- Therefore: $P\{S_8 = 2\} = \binom{8}{2}p^2q^6$.

Generalization: For exactly k successes in n trials:

$$P\{S_n = k\} = \binom{n}{k} p^k q^{n-k}, \quad k = 0, 1, \dots, n.$$

where the binomial coefficient is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n(n-1)\cdots(n-k+1)}{k!}.$$

4.4 Binomial Distribution

Definition 4.5 (Binomial Distribution). A random variable Z with values $\{0, 1, 2, \dots, n\}$ has the **Binomial Distribution** if

$$P\{Z = k\} = \binom{n}{k} p^k q^{n-k}, \quad k = 0, 1, \dots, n,$$

where $p \in (0, 1)$ and $q = 1 - p$. We write $Z \sim \text{Binom}(n, p)$.

In particular, $S_n \sim \text{Binom}(n, p)$.

Theorem 4.6 (Sanity Check: Probabilities Sum to 1).

$$\sum_{k=0}^n P\{Z = k\} = \sum_{k=0}^n \binom{n}{k} p^k q^{n-k} = (p + q)^n = 1^n = 1. \quad \checkmark$$

(Binomial theorem: $(p + q)^n = \sum_{k=0}^n \binom{n}{k} p^k q^{n-k}$.)

4.5 Expected Number of Successes

Theorem 4.7 ($E[S_n] = np$). If $S_n \sim \text{Binom}(n, p)$, then $E[S_n] = np$.

Proof via Indicator Functions (simpler). Write $S_n = X_1 + X_2 + \dots + X_n$ where each X_i is a Bernoulli(p) r.v. By linearity of expectation:

$$E[S_n] = E[X_1] + E[X_2] + \dots + E[X_n] = p + p + \dots + p = np.$$

□

Proof via Algebra (alternative).

$$\begin{aligned} E[S_n] &= \sum_{k=1}^n k \binom{n}{k} p^k q^{n-k} \quad (\text{term } k = 0 \text{ vanishes}) \\ &= \sum_{k=1}^n n \binom{n-1}{k-1} p^k q^{n-k} \quad (\text{since } k \binom{n}{k} = n \binom{n-1}{k-1}) \\ &= np \sum_{\ell=0}^{n-1} \binom{n-1}{\ell} p^\ell q^{n-1-\ell} \quad (\text{substituting } \ell = k - 1) \\ &= np \cdot (p + q)^{n-1} = np. \end{aligned}$$

□

4.6 Successes Between Trials

Example 4.8 (Successes between trials 10 and 21). Let $Z = S_{21} - S_9 = \sum_{k=10}^{21} X_k$. Since $X_{10}, X_{11}, \dots, X_{21}$ are i.i.d. Bernoulli(p) RVs, they have the same distribution as $X_1, \dots, X_{12}$. Therefore:

$$S_{21} - S_9 \sim \text{Binom}(12, p).$$

4.7 Incremental Successes (Independence of Non-Overlapping Intervals)

Example 4.9 ($P\{S_3 = 2, S_7 = 3\}$). • S_3 and S_7 are **not** independent (since S_7 depends on trials 1–3 which determine S_3).

- Rewrite: $\{S_3 = 2, S_7 = 3\} = \{S_3 = 2, S_7 - S_3 = 1\}$.
- $S_3 \sim \text{Binom}(3, p)$ and $S_7 - S_3 \sim \text{Binom}(4, p)$, and they **are** independent (non-overlapping intervals).
- Therefore:

$$P\{S_3 = 2, S_7 = 3\} = P\{S_3 = 2\} \cdot P\{S_7 - S_3 = 1\} = \binom{3}{2} p^2 q \cdot \binom{4}{1} p q^3.$$

5 Applying the Slides to HW3 Q4

This section shows how the tools above solve each part of Q4.

5.1 Part (a): Expected Number of Hookups for a Given Person

Fix a person i . There are $n - 1$ other people; for each person $j \neq i$, independently the website flips a coin with $P\{\text{heads}\} = p$.

Define the indicator for person i being matched with person j :

$$\mathbf{1}_{ij} = \begin{cases} 1, & \text{if persons } i \text{ and } j \text{ are matched} \\ 0, & \text{otherwise} \end{cases}$$

with $E[\mathbf{1}_{ij}] = P\{\text{match}\} = p$.

The total hookups for person i is:

$$H_i = \sum_{j \neq i} \mathbf{1}_{ij}.$$

By linearity of expectation:

$$E[H_i] = \sum_{j \neq i} E[\mathbf{1}_{ij}] = (n - 1) \cdot p.$$

5.2 Part (b): Probability of More than One Hookup

The hookups for person i are $H_i = \sum_{j \neq i} \mathbf{1}_{ij}$, where the $\mathbf{1}_{ij}$ are independent Bernoulli(p) RVs (independent by assumption). Therefore:

$$H_i \sim \text{Binom}(n - 1, p).$$

$$\begin{aligned} P\{H_i > 1\} &= 1 - P\{H_i = 0\} - P\{H_i = 1\} \\ &= 1 - \binom{n-1}{0} p^0 q^{n-1} - \binom{n-1}{1} p^1 q^{n-2} \\ &= \boxed{1 - q^{n-1} - (n-1)p q^{n-2}}, \end{aligned}$$

where $q = 1 - p$.

5.3 Part (c): Expected Number of Love Triangles

A love triangle is a set $\{i, j, k\}$ of three people who are all mutually matched.

Step 1: Define an indicator for each triple. For every unordered triple $\{i, j, k\}$, define:

$$\mathbf{1}_{ijk} = \begin{cases} 1, & \text{if all three pairs } (i, j), (j, k), (i, k) \text{ are matched} \\ 0, & \text{otherwise} \end{cases}$$

Step 2: Compute $E[\mathbf{1}_{ijk}]$. The three matches are independent (separate coin flips), so:

$$E[\mathbf{1}_{ijk}] = P\{\text{all three pairs matched}\} = p \cdot p \cdot p = p^3.$$

Step 3: Count all possible triples. The number of ways to choose 3 people from n is $\binom{n}{3} = \frac{n(n-1)(n-2)}{6}$.

Step 4: Apply linearity of expectation.

$$E[\text{total love triangles}] = \sum_{\{i,j,k\}} E[\mathbf{1}_{ijk}] = \binom{n}{3} \cdot p^3 = \boxed{\frac{n(n-1)(n-2)}{6} \cdot p^3}.$$